

10-2.1 Minkowski distance

$$Q \quad x = (1, 2, 3)$$

$$x' = (4, 6, 8)$$

Calculate MD ; a) $q=1$ (b) $q=2$ (c) $q=3$

$$d(x, x') = \left(\sum_{k=1}^d |x_k - x'_k|^q \right)^{\frac{1}{q}}, \quad q \geq 1$$

Solution (a) $q=1$

$$d_1(x, x') = \sum_{k=1}^3 |x_k - x'_k|$$

$$= |1-4| + |2-6| + |3-8|$$

$$= 3 + 4 + 5$$

$$= 12$$

(b) $q=2$

$$d_2(x, x') = \sqrt{\sum_{k=1}^3 (x_k - x'_k)^2}$$

$$= \sqrt{(1-4)^2 + (2-6)^2 + (3-8)^2}$$

$$= \sqrt{9 + 16 + 25}$$

$$= \sqrt{50}$$

$$= 5\sqrt{2}$$

$$(c) \quad q=3$$

$$d_3(x, x') = \left(\sum_{k=1}^3 |x_k - x'_k|^3 \right)^{\frac{1}{3}}$$

$$= \sqrt[3]{3^3 + 4^3 + 5^3}$$

$$= \sqrt[3]{27 + 64 + 125}$$

$$= \sqrt[3]{216}$$

$$= 6$$